RIVER VALLEY HIGH SCHOOL
YEAR 6 PRELIMINARY EXAMINATION II

CANDIDATE NAME

CENTRE NUMBER | INDEX NUMBER | CLASS

H2 BIOLOGY **9648/03**

Applications Paper and Planning Question **18 September 2014**

Paper 3 **2 hours**

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.
The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together. The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	/ 12
2	/ 13
3	/ 15
4 (SPA)	/ 12
5	/ 20
Total	

This Question Paper consists of **16** printed pages.

Answer **all** questions.

1. The Atlantic salmon, *Salmo salar*, is a fish in the family Salmonidae, which is found in the northern Atlantic Ocean. It has great commercial value in the food industry, and can be marketed as whole or processed. Its popularity has led to rising substitution with other non-salmon species.

A real-time Polymerase Chain Reaction (PCR) assay was developed for the identification of *Salmon salar*, allowing the detection of fraudulent or unintentional mislabelling of this species. This method can be applied to all fresh, frozen and processed products, including those that undergone intensive processes of transformation.

In this method, a pair of primers (forward and reverse primers) is designed to amplify the *ITS1* gene, a gene unique to the *Salmo* genus, during PCR. A radioactive DNA probe that would bind specifically to the *ITS1* gene is also designed.

Fig. 1.1 shows the binding positions of primers and DNA probe relative to the *ITS1* gene.

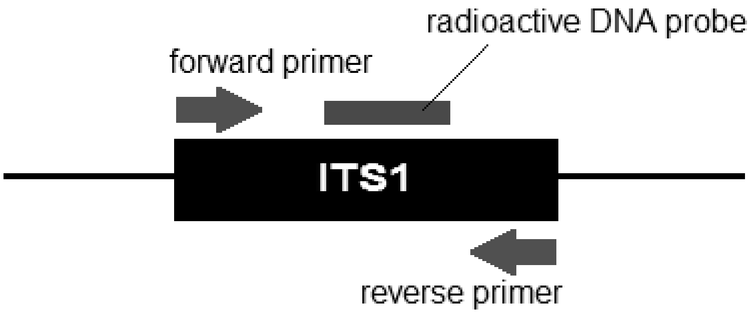

Fig. 1.1

Products labelled as 'Atlantic salmon' were bought from various markets and their DNA subjected to the PCR assay and Southern blotting.

The resulting autoradiogram is shown in **Fig. 1.2**.

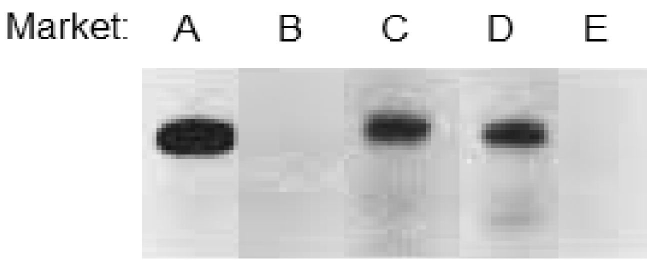

Fig. 1.2

1. **(a)** Describe how PCR amplifies the *ITS1* gene. [3]

(b) Explain why a pair of primers is needed for PCR. [2]

(c) Identify and explain which market(s) does not sell fish from the *Salmo* genus. [3]

1. **Fig.1.3** shows the amount of *ITSR* gene during PCR.

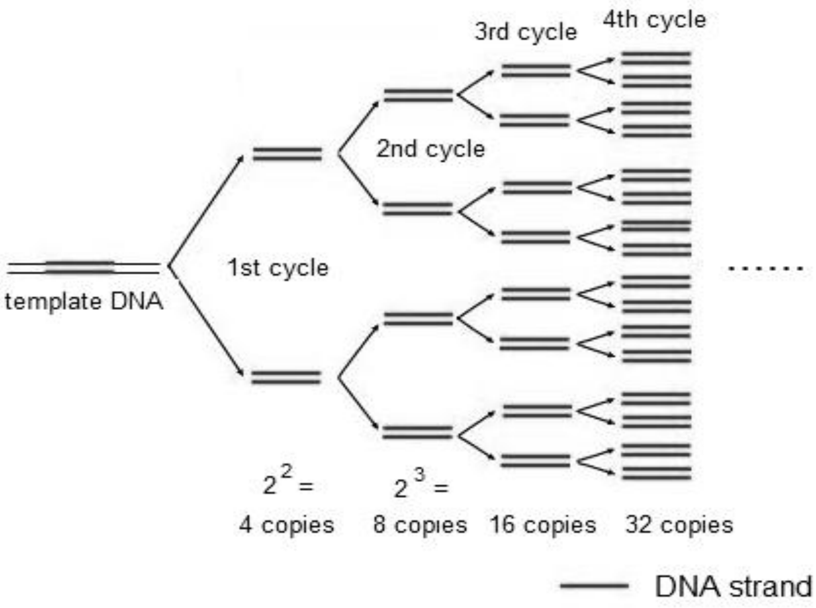

Fig. 1.3

(d) Indicate the number of *ITS1* genes present at the end of 18 cycles. Show **all** your working clearly. [1]

______________*ITS1* genes

(e) Discuss the limitations of using PCR for the authentication of *Salmo* genus. [2]

____

____

____

____

(f) Describe how the product synthesised by PCR is similar to that by DNA replication in eukaryotes. [1]

____

____

[Total: 12]

2. Adenosine deaminase (ADA) deficiency is an autosomal recessive metabolic disorder that causes immunodeficiency. Affected individuals are unable to produce the enzyme ADA. Without this enzyme, T lymphocytes cell count is compromised, and the immune system cannot defend against pathogens.

Gene therapy, first conceptualised in 1972, is a possible course of treatment for ADA deficient individuals. Although early clinical failures led to many dismissing gene therapy, recent clinical successes have bolstered new hope in this approach.

Fig. 2.1 shows the processes involved in the gene therapy treatment of ADA deficiency.

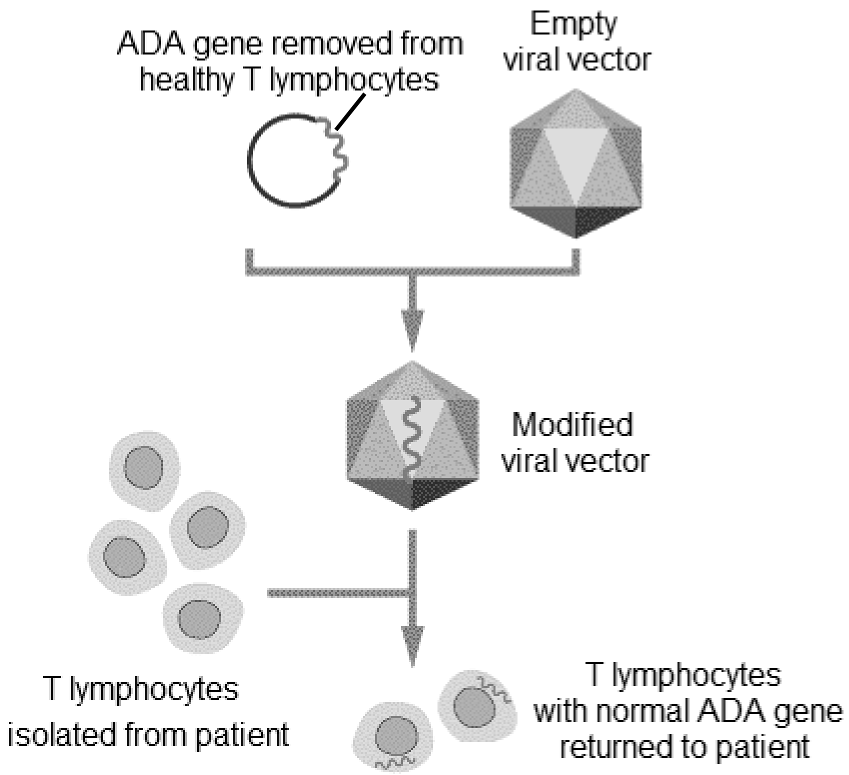

Image adapted from Klug & Cummings (1997)

Fig. 2.1

(a) Suggest why the early clinical trials of this treatment approach failed. [2]

2. **(b)** Explain the benefits of using viral vector in gene therapy. [2]

(c) Explain why individuals treated by this method of gene therapy would still have ADA-deficient children. [2]

Nine ADA deficient patients were administered with different doses of the modified viral vectors and empty viral vectors. Efficacy of the gene transfer was determined six days after the administration of respective vector. The results are shown in **Table 2.1** below.

Table 2.1

Patient	Gender	Age / years	Dose / pfu	Presence of ADA mRNA	
				Treated with empty viral vector	Treated with modified viral vector
1	F	25	2×10^7	No	No
2	M	44	2×10^7	No	No
3	M	23	2×10^8	No	No
4	M	21	2×10^8	No	No
5	F	34	2×10^9	No	Yes
6	M	40	2×10^9	No	No
7	F	19	2×10^9	No	Yes
8	F	33	2×10^{10}	No	Yes
9	M	22	2×10^{10}	No	No

2. **(d)** Describe the main trends shown in **Table 2.1**. [2]

(e) Suggest why patients are also treated with empty viral vectors. [1]

Another approach for the treatment of ADA deficiency involves bone marrow transplant from a genetically-matched donor. This, unlike the approach shown in **Fig. 2.1**, can provide a permanent cure with a single treatment.

(f) **(i)** Suggest why bone marrow transplant is obtained from a genetically-matched donor. [1]

(ii) Explain why a single bone marrow transplant can provide a permanent cure for ADA deficiency. [3]

[Total: 13]

3. Stem borers are insects that feed on the leaves and reside in the stem of plants, causing damage to many crops such as maize. An insecticidal toxin (Bt toxin) extracted from the bacterium, *Bacillus thuringiensis*, is found to be able to kill the stem borers but is harmless to other animals, including humans. The Bt toxin was used as an active ingredient in pesticides spray commonly used by farmers. However, there is limited success in completely killing all the stem borers on the plant.

In recent years, maize plant has been genetically modified to express a gene for Bt toxin. Tiny beads of tungsten were coated with plasmids containing the Bt gene and fired at very high speed at the maize cells to form Bt maize.

(a) Suggest why pesticides containing Bt toxin cannot effectively eliminate the stem borers from the plant. [1]

____

____

(b) Explain how Bt gene is inserted into the plasmid. [4]

____

____

____

____

____

____

____

____

(c) Explain the benefit of growing Bt maize. [1]

____

____

3. A study was carried out to see which types of Bt maize and their protein products would be most efficient against three species of stem borer. The stem borers were allowed to feed on nine types of maize (**A–I**), modified with Bt genes. The graph below shows the leaf areas damaged by the stem borers after feeding on maize leaves for five days.

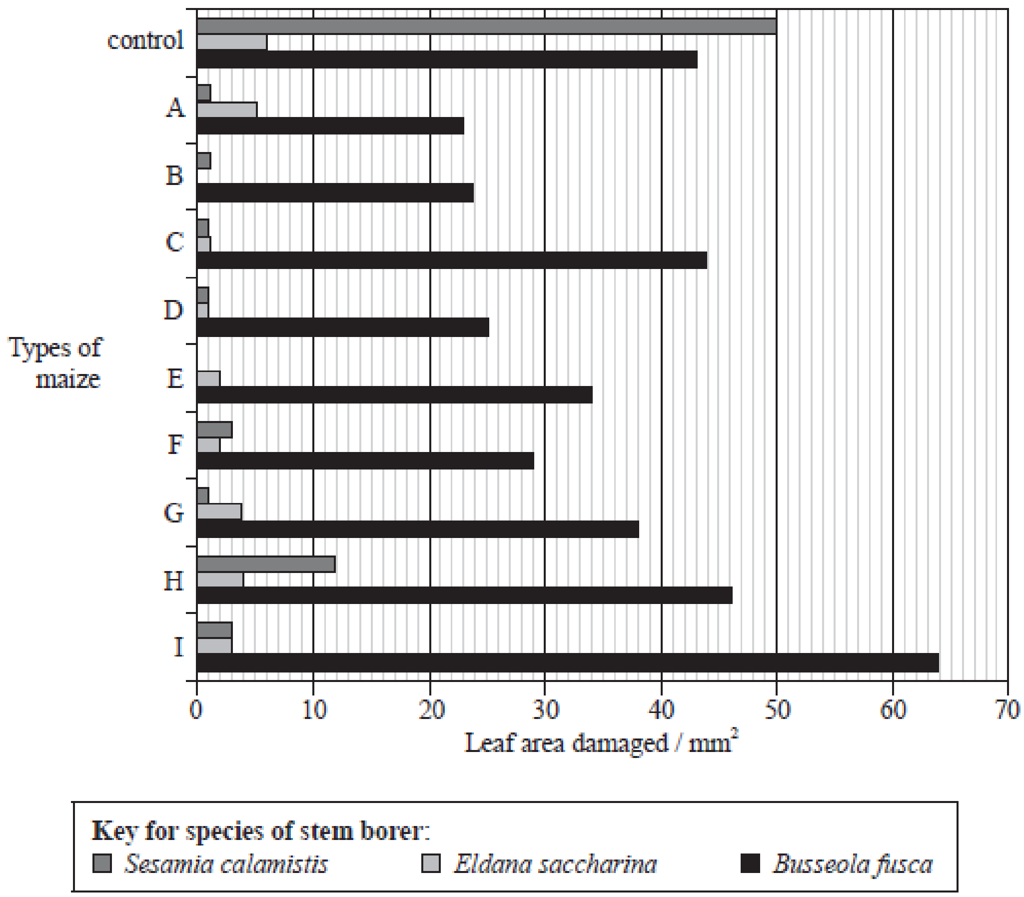

Image adapted from Mugo et al., 2005

Fig. 3.1

(d) With reference to **Fig. 3.1**,

(i) identity which species of stem borer was most successfully controlled by the Bt maize; [1]

(ii) explain your answer in **(d)(i)**. [2]

---

3. Some stem borers are resistant to Bt toxin. The Bt-resistant stem borers were used to find out the effect of Bt maize on the food chains. The stem borers may be parasitized by a wasp, which lays its egg inside them. The developing wasp larvae feed on the tissues of the stem borers and eventually emerge as adult wasps, killing the stem borers.

Groups of Bt-susceptible and Bt-resistant stem borers that were parasitised by wasps were fed either leaves of a Bt maize or leaves of a non-Bt maize. The percentage of parasitised stem borers from which the adult wasps emerged was then recorded.

The results of the investigation are shown in **Fig. 3.2**.

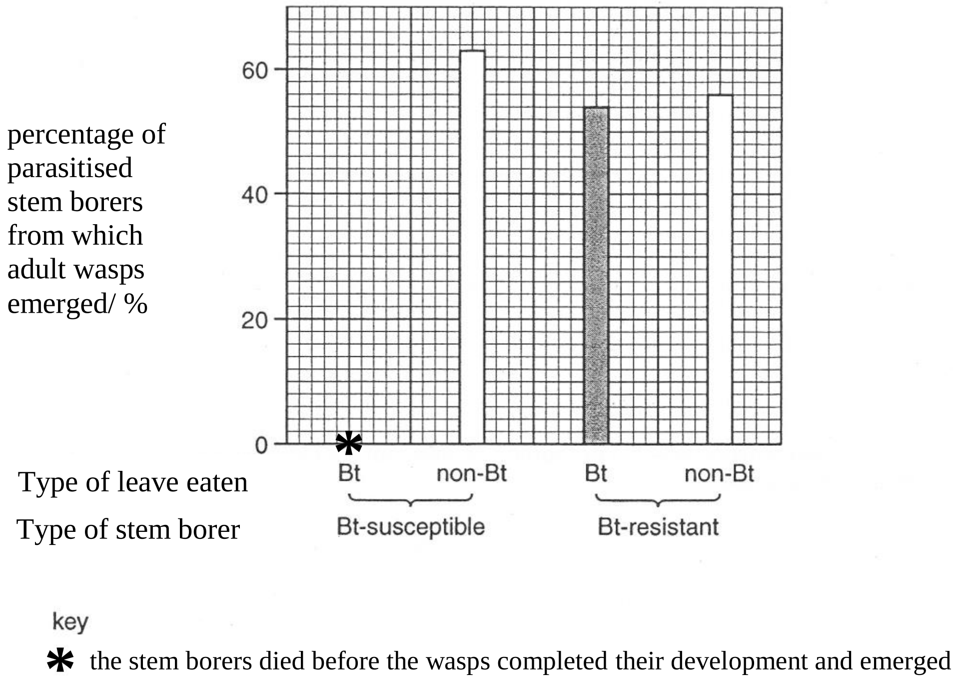

Image adapted from University of Cambridge Local Examinations, 2011

Fig.3.2

(e) With reference to **Fig. 3.2**,

(i) distinguish the effect of growing Bt maize and non-Bt maize on the wasp population. [3]

____

____

____

____

____

____

3. (e) (ii) describe how the results prove that Bt toxin kills wasp larvae. [1]

(f) Suggest the ecological implications of growing Bt maize. [2]

[Total: 15]

4. Planning Question

Cellular respiration is a process in which cells produce the energy they need for survival. In the presence of oxygen, most organisms undergo cellular respiration to break down sugars and store the energy in the form of adenosine triphosphate (ATP).

You are required to plan, but **not** carry out, an investigation into the effect of the type of carbohydrates on rate of respiration in yeast.

Fig. 4.1 shows a datalogger with carbon dioxide sensor probe that can be used to measure the concentration of carbon dioxide produced.

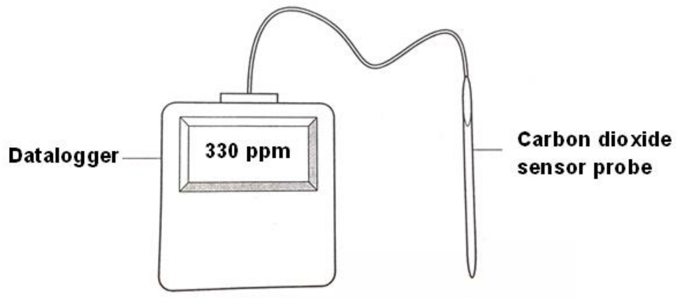

Fig. 4.1

Your planning must be based on the assumption that you have been provided with the following equipment and materials, which you must use:

- yeast suspension,
- 1% starch suspension,
- 1% sucrose solution,
- 10% glucose solution,
- rubber bung with an opening for probe
- Datalogger with carbon dioxide sensor probe
- timer, e.g. stopwatch,
- thermometer,
- a variety of different sized beakers, measuring cylinders or syringes for measuring volumes.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

Free Response Question

Write your answers on the separate answer paper provided.

Your answers:
- should be illustrated by large, clearly labelled diagrams, where appropriate.
- must be in continuous prose, where appropriate.
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL** return is necessary if you have not attempted this section.

5. **(a)** Micropropagation refers to the use of tissue culture techniques to propagate plants. Describe how this is carried out. [9]

(b) *"The Human Genome Project (HGP) was one of the great feats of exploration in history… (giving) the complete genetic blueprint for building a human being" – Human Genome Research Institute.*

Discuss the main objectives of this project. [4]

(c) Using a named example, explain how Restriction Fragment Length Polymorphism (RFLP) facilitates disease detection. [7]

[Total: 20]